

```
In [ ]: !git clone https://github.com/NTHung2034/PP-Final_Project.git
        %cd PP-Final_Project
```

```
Cloning into 'PP-Final_Project'...
remote: Enumerating objects: 1211, done.
remote: Counting objects: 100% (124/124), done.
remote: Compressing objects: 100% (88/88), done.
remote: Total 1211 (delta 67), reused 72 (delta 35), pack-reused 1087 (from 1)
Receiving objects: 100% (1211/1211), 106.25 MiB | 18.43 MiB/s, done.
Resolving deltas: 100% (640/640), done.
/content/PP-Final_Project
```

```
In [ ]: !bash ./scripts/download_cifar10.sh
```

```
Downloading CIFAR-10 dataset...
Data directory: ./data
--2025-12-13 13:15:13-- https://www.cs.toronto.edu/~kriz/cifar-10-binary.tar.gz
Resolving www.cs.toronto.edu (www.cs.toronto.edu)... 128.100.3.30
Connecting to www.cs.toronto.edu (www.cs.toronto.edu)|128.100.3.30|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 170052171 (162M) [application/x-gzip]
Saving to: 'cifar-10-binary.tar.gz'

cifar-10-binary.tar 100%[=====>] 162.17M 17.0MB/s in 11s

2025-12-13 13:15:25 (15.0 MB/s) - 'cifar-10-binary.tar.gz' saved [170052171/170052171]

CIFAR-10 dataset downloaded and extracted to ./data
Files:
-rw-r--r-- 1 2156 1103 30730000 Jun 4 2009 data_batch_1.bin
-rw-r--r-- 1 2156 1103 30730000 Jun 4 2009 data_batch_2.bin
-rw-r--r-- 1 2156 1103 30730000 Jun 4 2009 data_batch_3.bin
-rw-r--r-- 1 2156 1103 30730000 Jun 4 2009 data_batch_4.bin
-rw-r--r-- 1 2156 1103 30730000 Jun 4 2009 data_batch_5.bin
-rw-r--r-- 1 2156 1103 30730000 Jun 4 2009 test_batch.bin
```

```
In [ ]: !mkdir build
        %cd build
```

```
/content/PP-Final_Project/build
```

```
In [ ]: %cd /content/PP-Final_Project/build
        !cmake .. -DENABLE_CUDA=ON
        # compile
        !make
```

```

/content/PP-Final_Project/build
-- The CXX compiler identification is GNU 11.4.0
-- The C compiler identification is GNU 11.4.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Found CUDAToolkit: /usr/local/cuda/targets/x86_64-linux/include (found version "12.5.82")
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Success
-- Found Threads: TRUE
-- CUDA Toolkit found - GPU support enabled
-- The CUDA compiler identification is NVIDIA 12.5.82 with host compiler GNU 11.4.0
-- Detecting CUDA compiler ABI info
-- Detecting CUDA compiler ABI info - done
-- Check for working CUDA compiler: /usr/local/cuda/bin/nvcc - skipped
-- Detecting CUDA compile features
-- Detecting CUDA compile features - done
-- GPU Naive targets: train_gpu_naive, train_extract_naive, test_layers_gpu_naive, test_autoenc
oder_gpu_naive
-- GPU Optimized v1 targets: train_gpu_opt_v1, train_extract_opt_v1
-- GPU Optimized v2 targets: train_gpu_opt_v2, train_extract_opt_v2
-- Configuring done (3.8s)
-- Generating done (0.0s)
-- Build files have been written to: /content/PP-Final_Project/build
[ 1%] Building CXX object CMakeFiles/test_dataloader.dir/tests/test_dataloader.cpp.o
[ 2%] Building CXX object CMakeFiles/test_dataloader.dir/src/data/cifar10_loader.cpp.o
[ 3%] Linking CXX executable test_dataloader
[ 3%] Built target test_dataloader
[ 4%] Building CXX object CMakeFiles/test_layers_cpu.dir/tests/tests_cpu/test_layers.cpp.o
[ 5%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/conv2d_cpu.cpp.o
[ 6%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/relu_cpu.cpp.o
[ 7%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/maxpool_cpu.cpp.o
[ 8%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/upsample_cpu.cpp.o
[ 9%] Linking CXX executable test_layers_cpu
[ 9%] Built target test_layers_cpu
[10%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/tests/tests_cpu/test_autoencod
er.cpp.o
[11%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/conv2d_cpu.cpp.o
[12%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/relu_cpu.cpp.o
[13%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/maxpool_cpu.cpp.o
[14%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/upsample_cpu.cpp.o
[15%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/models/autoencoder_cpu.cpp.o
[16%] Linking CXX executable test_autoencoder_cpu
[16%] Built target test_autoencoder_cpu
[17%] Building CXX object CMakeFiles/train_cpu.dir/src/train_cpu.cpp.o
[18%] Building CXX object CMakeFiles/train_cpu.dir/src/data/cifar10_loader.cpp.o
[19%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/conv2d_cpu.cpp.o
[20%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/relu_cpu.cpp.o
[21%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/maxpool_cpu.cpp.o
[22%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/upsample_cpu.cpp.o
[23%] Building CXX object CMakeFiles/train_cpu.dir/src/models/autoencoder_cpu.cpp.o
[24%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/logger.cpp.o
[25%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/image_utils.cpp.o
[26%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/memory_tracker.cpp.o
[27%] Linking CXX executable train_cpu

```

```
[ 27%] Built target train_cpu
[ 28%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/extract_features_cpu.cpp.o
[ 29%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/data/cifar10_loader.cpp.o
[ 30%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/conv2d_cpu.cpp.o
[ 31%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/relu_cpu.cpp.o
[ 32%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/maxpool_cpu.cpp.o
[ 33%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/upsample_cpu.cpp.o
[ 34%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/models/autoencoder_cpu.cpp.o
[ 35%] Linking CXX executable extract_features_cpu
[ 35%] Built target extract_features_cpu
[ 36%] Building CXX object CMakeFiles/train_svm_cpu.dir/src/models_svm/train_svm_cpu.cpp.o
[ 37%] Building CXX object CMakeFiles/train_svm_cpu.dir/src/data/cifar10_loader.cpp.o
[ 38%] Building CXX object CMakeFiles/train_svm_cpu.dir/external/svm.cpp.o
[ 39%] Linking CXX executable train_svm_cpu
[ 39%] Built target train_svm_cpu
[ 40%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/train_gpu_naive.cu.o
[ 41%] Building CXX object CMakeFiles/train_gpu_naive.dir/src/data/cifar10_loader.cpp.o
[ 42%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/conv2d_gpu_naive.cu.o
[ 43%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/maxpool_gpu_naive.cu.o
[ 44%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/relu_gpu_naive.cu.o
[ 45%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/upsample_gpu_naive.cu.o
[ 46%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/mse_loss_gpu_naive.cu.o
[ 47%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/models/autoencoder_gpu_naive.cu.o
[ 48%] Linking CUDA device code CMakeFiles/train_gpu_naive.dir/cmake_device_link.o
[ 49%] Linking CXX executable train_gpu_naive
[ 49%] Built target train_gpu_naive
[ 50%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/models_svm/train_extract_naive.cu.o
[ 51%] Building CXX object CMakeFiles/train_extract_naive.dir/src/data/cifar10_loader.cpp.o
[ 52%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/conv2d_gpu_naive.cu.o
[ 53%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/maxpool_gpu_naive.cu.o
[ 54%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/relu_gpu_naive.cu.o
[ 55%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/upsample_gpu_naive.cu.o
[ 56%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/mse_loss_gpu_naive.cu.o
[ 57%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/models/autoencoder_gpu_naive.cu.o
[ 58%] Linking CUDA device code CMakeFiles/train_extract_naive.dir/cmake_device_link.o
[ 59%] Linking CXX executable train_extract_naive
[ 59%] Built target train_extract_naive
[ 60%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/train_gpu_opt_v1.cu.o
[ 61%] Building CXX object CMakeFiles/train_gpu_opt_v1.dir/src/data/cifar10_loader.cpp.o
[ 62%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/conv2d_gpu_opt_v1.cu.o
[ 63%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/relu_gpu_opt_v1.cu.o
[ 64%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/maxpool_gpu_opt_v1.cu.o
[ 65%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/upsample_gpu_opt_v1.cu.o
[ 66%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/mse_loss_gpu_opt_v1.cu.o
```

```

[ 67%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/models/autoencoder_gpu_opt_v1.cu.o
[ 68%] Linking CUDA device code CMakeFiles/train_gpu_opt_v1.dir/cmake_device_link.o
[ 69%] Linking CXX executable train_gpu_opt_v1
[ 69%] Built target train_gpu_opt_v1
[ 70%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/models_svm/train_extract_opt_v1.cu.o
[ 71%] Building CXX object CMakeFiles/train_extract_opt_v1.dir/src/data/cifar10_loader.cpp.o
[ 72%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/conv2d_gpu_opt_v1.cu.o
[ 73%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/relu_gpu_opt_v1.cu.o
[ 74%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/maxpool_gpu_opt_v1.cu.o
[ 75%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/upsample_gpu_opt_v1.cu.o
[ 76%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/mse_loss_gpu_opt_v1.cu.o
[ 77%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/models/autoencoder_gpu_opt_v1.cu.o
[ 78%] Linking CUDA device code CMakeFiles/train_extract_opt_v1.dir/cmake_device_link.o
[ 79%] Linking CXX executable train_extract_opt_v1
[ 79%] Built target train_extract_opt_v1
[ 80%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/train_gpu_opt_v2.cu.o
[ 81%] Building CXX object CMakeFiles/train_gpu_opt_v2.dir/src/data/cifar10_loader.cpp.o
[ 82%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/conv2d_gpu_opt_v2.cu.o
[ 83%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/relu_gpu_opt_v2.cu.o
[ 84%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/maxpool_gpu_opt_v2.cu.o
[ 85%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/upsample_gpu_opt_v2.cu.o
[ 86%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/mse_loss_gpu_opt_v2.cu.o
[ 87%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/models/autoencoder_gpu_opt_v2.cu.o
[ 88%] Linking CUDA device code CMakeFiles/train_gpu_opt_v2.dir/cmake_device_link.o
[ 89%] Linking CXX executable train_gpu_opt_v2
[ 89%] Built target train_gpu_opt_v2
[ 90%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/models_svm/train_extract_opt_v2.cu.o
[ 91%] Building CXX object CMakeFiles/train_extract_opt_v2.dir/src/data/cifar10_loader.cpp.o
[ 92%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/conv2d_gpu_opt_v2.cu.o
[ 93%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/relu_gpu_opt_v2.cu.o
[ 94%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/maxpool_gpu_opt_v2.cu.o
[ 95%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/upsample_gpu_opt_v2.cu.o
[ 96%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/mse_loss_gpu_opt_v2.cu.o
[ 97%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/models/autoencoder_gpu_opt_v2.cu.o
[ 98%] Linking CUDA device code CMakeFiles/train_extract_opt_v2.dir/cmake_device_link.o
[100%] Linking CXX executable train_extract_opt_v2
[100%] Built target train_extract_opt_v2

```

```
In [ ]: !./train_gpu_opt_v1
```

CIFAR-10 AUTOENCODER - GPU OPTIMIZED v1 (Memory Optimization)

Using GPU: Tesla T4
Compute Capability: 7.5
Total Global Memory: 15095 MB

VRAM usage: 102 MB

Loading CIFAR-10 training data...
Loaded data_batch_1.bin
Loaded data_batch_2.bin
Loaded data_batch_3.bin
Loaded data_batch_4.bin
Loaded data_batch_5.bin
Training data loaded: 50000 images

GPU OPTIMIZED v1 AUTOENCODER TRAINING

(Memory Pool + Shared Memory Tiling + Constant Memory)

Configuration:

Batch size: 64
Epochs: 20
Learning rate: 0.001
Training samples: 50000
Save directory: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1

VRAM usage: 490 MB

Batches per epoch: 781

Epoch [1/20]	Loss: 0.057110	Time: 180.25s	Avg: 180.25s/epoch
Epoch [2/20]	Loss: 0.035706	Time: 185.55s	Avg: 182.90s/epoch
Epoch [3/20]	Loss: 0.028188	Time: 185.76s	Avg: 183.85s/epoch
Epoch [4/20]	Loss: 0.024393	Time: 185.59s	Avg: 184.28s/epoch
Epoch [5/20]	Loss: 0.022084	Time: 185.44s	Avg: 184.51s/epoch
Epoch [6/20]	Loss: 0.020665	Time: 185.34s	Avg: 184.65s/epoch
Epoch [7/20]	Loss: 0.019724	Time: 185.46s	Avg: 184.77s/epoch
Epoch [8/20]	Loss: 0.019098	Time: 185.40s	Avg: 184.85s/epoch
Epoch [9/20]	Loss: 0.018760	Time: 185.45s	Avg: 184.91s/epoch
Epoch [10/20]	Loss: 0.018392	Time: 185.50s	Avg: 184.97s/epoch
Epoch [11/20]	Loss: 0.018157	Time: 185.65s	Avg: 185.03s/epoch
Epoch [12/20]	Loss: 0.017828	Time: 185.57s	Avg: 185.08s/epoch
Epoch [13/20]	Loss: 0.017546	Time: 185.58s	Avg: 185.12s/epoch
Epoch [14/20]	Loss: 0.017398	Time: 185.56s	Avg: 185.15s/epoch
Epoch [15/20]	Loss: 0.017187	Time: 185.70s	Avg: 185.18s/epoch
Epoch [16/20]	Loss: 0.017002	Time: 185.63s	Avg: 185.21s/epoch
Epoch [17/20]	Loss: 0.016784	Time: 185.80s	Avg: 185.25s/epoch
Epoch [18/20]	Loss: 0.016548	Time: 185.66s	Avg: 185.27s/epoch
Epoch [19/20]	Loss: 0.016408	Time: 185.56s	Avg: 185.29s/epoch
Epoch [20/20]	Loss: 0.016209	Time: 185.65s	Avg: 185.30s/epoch

TRAINING COMPLETED

Total training time: 3706.07s
Average time per epoch: 185.30s
Initial loss: 0.057110

Final loss: 0.016209
Loss reduction: 71.617386%

=====

Training summary saved to: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/training_summary.txt
Final VRAM usage: 492 MB

TRAINING COMPLETED!

In []: !./train_extract_opt_v1

CIFAR-10 FEATURE EXTRACTION - GPU OPT V1 (PHASE 4)

=== Step 1: Loading Dataset ===

Loading CIFAR-10 training data...

Loaded data_batch_1.bin

Loaded data_batch_2.bin

Loaded data_batch_3.bin

Loaded data_batch_4.bin

Loaded data_batch_5.bin

Training data loaded: 50000 images

Loading CIFAR-10 test data...

Test data loaded: 10000 images

Training images: 50000

Test images: 10000

=== Step 2: Loading Encoder Weights ===

Encoder weights loaded successfully

=== Step 3: Extracting Features ===

Train batch [50/500]

Train batch [100/500]

Train batch [150/500]

Train batch [200/500]

Train batch [250/500]

Train batch [300/500]

Train batch [350/500]

Train batch [400/500]

Train batch [450/500]

Train batch [500/500]

Test batch [20/100]

Test batch [40/100]

Test batch [60/100]

Test batch [80/100]

Test batch [100/100]

Feature extraction completed: 29.29s

=== Step 4: Saving Features to .bin Files ===

Saved train features: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/train_features.bin

Shape: [50000, 8192]

Saved train labels: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/train_labels.bin

Shape: [50000]

Saved test features: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/test_features.bin

Shape: [10000, 8192]

Saved test labels: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/test_labels.bin

Shape: [10000]

Saved metadata: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/metadata.txt

=== Summary ===

Feature extraction time: 29.29s

All features saved to: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features

FEATURE EXTRACTION COMPLETED!

```
In [ ]: %cd /content/PP-Final_Project/  
!python /content/PP-Final_Project/src/models_svm/cuml_svm_train.py opt_v1
```

/content/PP-Final_Project

```
=====
CIFAR-10 cudaML SVM Training with RBF Kernel [OPT_V1]
=====
```

Configuration:

Implementation: OPT_V1
Data directory: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features
Kernel: rbf
C: 10.0
Gamma: auto

=== Step 1: Loading Data ===

Loaded features shape: (50000, 8192)
Loaded labels shape: (50000,)
Loaded features shape: (10000, 8192)
Loaded labels shape: (10000,)
Train samples: 50000
Test samples: 10000

Normalizing features...

Features normalized (mean=0, std=1)

=== Step 2: Training SVM with cudaML ===

Initializing SVM classifier...

Training started...

[2025-12-13 14:20:40.112] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:40.984] [CUMML] [debug] SMO solver finished after 46 outer iterations, total inner 23029 iterations, and diff 0.000991
[2025-12-13 14:20:41.582] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:42.326] [CUMML] [debug] SMO solver finished after 57 outer iterations, total inner 31134 iterations, and diff 0.000997
[2025-12-13 14:20:42.829] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:43.424] [CUMML] [debug] SMO solver finished after 43 outer iterations, total inner 22923 iterations, and diff 0.000999
[2025-12-13 14:20:43.929] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:44.553] [CUMML] [debug] SMO solver finished after 46 outer iterations, total inner 25462 iterations, and diff 0.000976
[2025-12-13 14:20:45.058] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:45.588] [CUMML] [debug] SMO solver finished after 38 outer iterations, total inner 19778 iterations, and diff 0.000999
[2025-12-13 14:20:46.090] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:46.531] [CUMML] [debug] SMO solver finished after 32 outer iterations, total inner 16166 iterations, and diff 0.000985
[2025-12-13 14:20:47.049] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:47.637] [CUMML] [debug] SMO solver finished after 42 outer iterations, total inner 22234 iterations, and diff 0.000979
[2025-12-13 14:20:48.143] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:49.024] [CUMML] [debug] SMO solver finished after 66 outer iterations, total inner 38372 iterations, and diff 0.000996
[2025-12-13 14:20:49.527] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:50.211] [CUMML] [debug] SMO solver finished after 50 outer iterations, total inner 25266 iterations, and diff 0.000991
[2025-12-13 14:20:50.714] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:51.229] [CUMML] [debug] SMO solver finished after 36 outer iterations, total inner 17443 iterations, and diff 0.000987
[2025-12-13 14:20:51.732] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:52.334] [CUMML] [debug] SMO solver finished after 41 outer iterations, total inner 20026 iterations, and diff 0.000998
[2025-12-13 14:20:52.832] [CUMML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:53.320] [CUMML] [debug] SMO solver finished after 33 outer iterations, total inner

nner 15686 iterations, and diff 0.000997
[2025-12-13 14:20:53.818] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:54.343] [CUML] [debug] SMO solver finished after 35 outer iterations, total i
nner 16540 iterations, and diff 0.000992
[2025-12-13 14:20:54.846] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:55.348] [CUML] [debug] SMO solver finished after 33 outer iterations, total i
nner 15440 iterations, and diff 0.000993
[2025-12-13 14:20:55.843] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:56.423] [CUML] [debug] SMO solver finished after 40 outer iterations, total i
nner 19161 iterations, and diff 0.000990
[2025-12-13 14:20:56.929] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:57.608] [CUML] [debug] SMO solver finished after 48 outer iterations, total i
nner 24231 iterations, and diff 0.000999
[2025-12-13 14:20:58.117] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:20:59.205] [CUML] [debug] SMO solver finished after 78 outer iterations, total i
nner 38694 iterations, and diff 0.000997
[2025-12-13 14:20:59.717] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:00.797] [CUML] [debug] SMO solver finished after 82 outer iterations, total i
nner 45187 iterations, and diff 0.000978
[2025-12-13 14:21:01.301] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:02.444] [CUML] [debug] SMO solver finished after 87 outer iterations, total i
nner 53521 iterations, and diff 0.000997
[2025-12-13 14:21:02.946] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:04.011] [CUML] [debug] SMO solver finished after 79 outer iterations, total i
nner 44033 iterations, and diff 0.000995
[2025-12-13 14:21:04.515] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:05.441] [CUML] [debug] SMO solver finished after 67 outer iterations, total i
nner 38392 iterations, and diff 0.000996
[2025-12-13 14:21:05.943] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:06.844] [CUML] [debug] SMO solver finished after 66 outer iterations, total i
nner 35374 iterations, and diff 0.000999
[2025-12-13 14:21:07.354] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:07.886] [CUML] [debug] SMO solver finished after 37 outer iterations, total i
nner 19815 iterations, and diff 0.000981
[2025-12-13 14:21:08.390] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:08.949] [CUML] [debug] SMO solver finished after 38 outer iterations, total i
nner 18576 iterations, and diff 0.000998
[2025-12-13 14:21:09.452] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:10.481] [CUML] [debug] SMO solver finished after 75 outer iterations, total i
nner 41813 iterations, and diff 0.001000
[2025-12-13 14:21:10.982] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:12.392] [CUML] [debug] SMO solver finished after 104 outer iterations, total
inner 55908 iterations, and diff 0.000987
[2025-12-13 14:21:12.895] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:13.907] [CUML] [debug] SMO solver finished after 71 outer iterations, total i
nner 40382 iterations, and diff 0.000972
[2025-12-13 14:21:14.410] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:15.372] [CUML] [debug] SMO solver finished after 67 outer iterations, total i
nner 35458 iterations, and diff 0.000992
[2025-12-13 14:21:15.882] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:16.426] [CUML] [debug] SMO solver finished after 37 outer iterations, total i
nner 18356 iterations, and diff 0.000998
[2025-12-13 14:21:16.930] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:17.576] [CUML] [debug] SMO solver finished after 43 outer iterations, total i
nner 21415 iterations, and diff 0.000977
[2025-12-13 14:21:18.078] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:19.053] [CUML] [debug] SMO solver finished after 69 outer iterations, total i
nner 38788 iterations, and diff 0.000990
[2025-12-13 14:21:19.561] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:20.534] [CUML] [debug] SMO solver finished after 69 outer iterations, total i
nner 40418 iterations, and diff 0.000996
[2025-12-13 14:21:21.036] [CUML] [debug] Creating working set with 1024 elements

```
[2025-12-13 14:21:22.075] [CUML] [debug] SMO solver finished after 74 outer iterations, total i
nner 41166 iterations, and diff 0.000960
[2025-12-13 14:21:22.579] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:23.092] [CUML] [debug] SMO solver finished after 35 outer iterations, total i
nner 17739 iterations, and diff 0.000973
[2025-12-13 14:21:23.598] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:24.116] [CUML] [debug] SMO solver finished after 33 outer iterations, total i
nner 16460 iterations, and diff 0.000993
[2025-12-13 14:21:24.614] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:25.466] [CUML] [debug] SMO solver finished after 59 outer iterations, total i
nner 32380 iterations, and diff 0.000994
[2025-12-13 14:21:25.968] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:26.978] [CUML] [debug] SMO solver finished after 70 outer iterations, total i
nner 37004 iterations, and diff 0.000996
[2025-12-13 14:21:27.488] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:27.956] [CUML] [debug] SMO solver finished after 31 outer iterations, total i
nner 15186 iterations, and diff 0.000995
[2025-12-13 14:21:28.455] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:29.038] [CUML] [debug] SMO solver finished after 38 outer iterations, total i
nner 18459 iterations, and diff 0.000988
[2025-12-13 14:21:29.547] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:30.237] [CUML] [debug] SMO solver finished after 47 outer iterations, total i
nner 24742 iterations, and diff 0.000999
[2025-12-13 14:21:30.740] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:31.114] [CUML] [debug] SMO solver finished after 25 outer iterations, total i
nner 11979 iterations, and diff 0.000991
[2025-12-13 14:21:31.623] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:32.157] [CUML] [debug] SMO solver finished after 34 outer iterations, total i
nner 16926 iterations, and diff 0.000991
[2025-12-13 14:21:32.654] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:33.110] [CUML] [debug] SMO solver finished after 30 outer iterations, total i
nner 14887 iterations, and diff 0.000986
[2025-12-13 14:21:33.611] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:34.308] [CUML] [debug] SMO solver finished after 46 outer iterations, total i
nner 22800 iterations, and diff 0.000990
[2025-12-13 14:21:34.816] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:21:35.497] [CUML] [debug] SMO solver finished after 46 outer iterations, total i
nner 23948 iterations, and diff 0.000982
Training completed in 63.47s
```

=== Step 3: Evaluation ===

Evaluating on training set...

Processed 10000/50000 samples

Processed 20000/50000 samples

Processed 30000/50000 samples

Processed 40000/50000 samples

Processed 50000/50000 samples

Evaluating on test set...

Training Accuracy: 99.37%

Test Accuracy: 63.97%

Confusion matrix saved to: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_featur
es/confusion_matrix.png

=== Per-Class Accuracy ===

airplane: 69.00%

automobile: 73.60%

bird: 49.20%

cat: 51.70%

deer: 57.80%

dog: 53.90%

frog: 71.30%

horse: 67.60%
ship: 75.20%
truck: 70.40%

=== Classification Report ===

	precision	recall	f1-score	support
airplane	0.6873	0.6900	0.6886	1000
automobile	0.7518	0.7360	0.7438	1000
bird	0.5694	0.4920	0.5279	1000
cat	0.4042	0.5170	0.4537	1000
deer	0.6110	0.5780	0.5940	1000
dog	0.5650	0.5390	0.5517	1000
frog	0.7130	0.7130	0.7130	1000
horse	0.7056	0.6760	0.6905	1000
ship	0.7558	0.7520	0.7539	1000
truck	0.6895	0.7040	0.6967	1000
accuracy			0.6397	10000
macro avg	0.6453	0.6397	0.6414	10000
weighted avg	0.6453	0.6397	0.6414	10000

=== Step 4: Saving Model ===

Model saved to: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/cuml_svm_model.pkl

=====

TRAINING SUMMARY

=====

Configuration:

Implementation: OPT_V1
Kernel: rbf
C: 10.0
Gamma: auto

Training time: 63.47s
Training accuracy: 99.37%
Test accuracy: 63.97%

=====

=== Per-Class Accuracy ===

airplane: 69.00%
automobile: 73.60%
bird: 49.20%
cat: 51.70%
deer: 57.80%
dog: 53.90%
frog: 71.30%
horse: 67.60%
ship: 75.20%
truck: 70.40%

=== Classification Report ===

	precision	recall	f1-score	support
airplane	0.6873	0.6900	0.6886	1000
automobile	0.7518	0.7360	0.7438	1000
bird	0.5694	0.4920	0.5279	1000

cat	0.4042	0.5170	0.4537	1000
deer	0.6110	0.5780	0.5940	1000
dog	0.5650	0.5390	0.5517	1000
frog	0.7130	0.7130	0.7130	1000
horse	0.7056	0.6760	0.6905	1000
ship	0.7558	0.7520	0.7539	1000
truck	0.6895	0.7040	0.6967	1000
accuracy			0.6397	10000
macro avg	0.6453	0.6397	0.6414	10000
weighted avg	0.6453	0.6397	0.6414	10000

Training summary saved to: /content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/training_summary.txt

```
In [ ]: !zip -r models_v1.zip /content/PP-Final_Project/models
from google.colab import files
files.download("models_v1.zip")

adding: content/PP-Final_Project/models/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/training_summary.txt (deflated 43%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/conv3_w.bin (deflated 8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/conv3_b.bin (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/conv5_w.bin (deflated 8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/conv2_w.bin (deflated 9%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/conv2_b.bin (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/conv1_b.bin (deflated 1%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/conv1_w.bin (deflated 8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/training_summary.txt (deflated 65%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/test_labels.bin (deflated 49%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/test_features.bin (deflated 40%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/metadata.txt (deflated 24%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/confusion_matrix.png (deflated 16%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/train_labels.bin (deflated 51%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/cuml_svm_model.pkl (deflated 39%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/svm_features/train_features.bin (deflated 40%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/conv5_b.bin (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/conv4_w.bin (deflated 8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/conv4_b.bin (deflated 1%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights/epoch_metrics_cpu.txt (deflated 3%)
adding: content/PP-Final_Project/models/saved_weights/svm_results.txt (deflated 47%)
adding: content/PP-Final_Project/models/saved_weights/test_predictions.txt (deflated 65%)
adding: content/PP-Final_Project/models/saved_weights/cpu_final.bin (deflated 7%)
adding: content/PP-Final_Project/models/saved_weights/test_weights.bin (deflated 7%)
adding: content/PP-Final_Project/models/saved_weights/test_features_cpu.bin (deflated 50%)
adding: content/PP-Final_Project/models/saved_weights/train_features_cpu.bin (deflated 50%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_optimized/ (stored 0%)
```

